School of Computer Science and Artificial Intelligence

Lab Assignment # 23

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Task 1 – Project Proposal

- **Students choose a real-world problem to solve with an AI-assisted application (e.g., e-commerce billing system, health monitoring dashboard, student attendance tracker).**
- **Prepare a project proposal including:**

o Objective
o Features
o Tools/technologies to be used
o Expected output

Prompt:-
Create a simple project proposal for an AI-assisted application that solves a real-world problem. Include objective, key features, technologies used, and expected output. Keep the explanation clear, practical, and student-friendly. Make it suitable for a college capstone project.

Code:-

```
# Task - 01
# Create a simple project proposal for an AI-assisted application that solves a real-world problem. Include objective, key features, techn
"""# Project Proposal: AI-Assisted Personal Fitness Trainer
# Objective: To develop an AI-assisted application that provides personalized fitness training plans and real-time guidance to users based
on their goals, fitness level, and progress.
# Key Features:
# 1. Personalized Workout Plans: The application will generate customized workout plans based on user input such as age, weight, height,
fitness goals (e.g., weight loss, muscle gain), and preferred workout types (e.g., cardio, strength training).
# 2. Real-Time Guidance: The application will provide real-time feedback and guidance during workouts, ensuring users maintain proper
form and technique.
# 3. Progress Tracking: Users can track their progress over time, including workout history, improvements in strength and endurance, and
changes in body measurements.
# 4. Nutrition Recommendations: The application will offer personalized nutrition advice to complement the workout plans, helping users
achieve their fitness goals more effectively.
# Technologies Used:
# 1. Machine Learning: To analyze user data and generate personalized workout plans and nutrition recommendations.
# 2. Computer Vision: To provide real-time feedback on exercise form and technique using the user's device camera.
# 3. Mobile App Development: To create a user-friendly interface for both iOS and Android platforms.
# 4. Cloud Computing: To store user data securely and enable real-time processing of workout feedback.
```

Output:-
The expected output of the project is a fully functional AI-assisted personal fitness trainer application that can be used by individuals to achieve their fitness goals. The application will provide personalized workout plans, real-time guidance during exercises, and nutrition recommendations, ultimately helping users improve their fitness levels and overall health.

Justification:
This project addresses a common real-world problem of maintaining fitness and achieving health goals. By leveraging AI technologies, the application can provide personalized and effective solutions for users, making it a valuable tool for anyone looking to improve their fitness. Additionally, the project offers opportunities for students to learn and apply various technologies such as machine learning, computer vision, and mobile app development, making it an ideal choice for a college capstone project.

Task 2 – Application Design
• Use AI tools to assist in:
o Designing system architecture diagrams.
o Defining database schema (if needed).
o Creating wireframes/UI mockups for frontend design.

Prompt:-
Design the system architecture for the chosen application using simple explanations. Include frontend, backend, and database flow clearly. Provide a basic database schema and UI wireframe ideas. Keep the design easy to understand and implement.

Code:-

```
# Task - 02
# Design the system architecture for the chosen application using simple explanations. Include frontend, backend, and database flow clearly
"""
System Architecture for AI-Assisted Personal Fitness Trainer Application:
1. Frontend:
   - User Interface (UI): The frontend will consist of a mobile application interface that allows users to input their personal information
   fitness goals, and preferences. It will also display workout plans, real-time feedback during exercises, and progress tracking.
   - Technologies: React Native for cross-platform mobile app development.
2. Backend:
   - API Server: The backend will have an API server that handles requests from the frontend, processes user data, and interacts with the
   machine learning models and database.
   - Machine Learning Models: The backend will include machine learning models for generating personalized workout plans and nutrition
   recommendations based on user data.
   - Computer Vision Module: This module will process video input from the user's device camera to provide real-time feedback on exercise
   form and technique.
   - Technologies: Node.js for the API server, Python for machine learning and computer vision modules
3. Database:
   - The database will store user information, workout history, progress data, and nutrition recommendations.
    - Technologies: MongoDB for a flexible, document-oriented database.
Database Schema:
- Users Collection: { _id: ObjectId, name: String, age: Number, weight: Number, height: Number, fitnessGoals: [String],
preferredWorkoutTypes: [String] }
- Workouts Collection: { _id: ObjectId, userId: ObjectId, workoutPlan: [String], date: Date }
- Progress Collection: { _id: ObjectId, userId: ObjectId, workoutHistory: [String], strengthImprovements: [String], enduranceImprovements:
[String], bodyMeasurements: { weight: Number, waist: Number, hips: Number } }
UI Wireframe Ideas:
1. Home Screen: Displays a welcome message, user profile, and quick access to workout plans and progress tracking.
2. Workout Plan Screen: Allows users to input their fitness goals and preferences, and displays the generated workout plan with options to
start the workout and receive real-time feedback.
3. Progress Tracking Screen: Shows the user's workout history, improvements in strength and endurance, and changes in body measurements
over time, with visual graphs and charts for easy understanding.
```

Output:-
The expected output of the system architecture design is a clear and practical blueprint for developing the AI-assisted personal fitness trainer application. The architecture outlines the components and technologies needed for the frontend, backend, and database, along with a basic database schema and UI wireframe ideas. This design will guide the implementation of the application, ensuring that it meets the objectives and provides a user-friendly experience for individuals looking to improve their fitness levels.

Justification:
The proposed system architecture is designed to be modular and scalable, allowing for future enhancements such as additional features or increased user load. The use of React Native for the frontend ensures that the application can reach a wide audience across both iOS and Android platforms. The backend technologies chosen are well-suited for handling the processing requirements of machine learning and computer vision tasks, while MongoDB provides a flexible database solution that can easily accommodate changes in data structure as the application evolves. Overall, this architecture provides a solid foundation for developing a successful AI-assisted personal fitness trainer application.

Task 3 – AI-Assisted Code Development
• Implement the application in phases:
o Frontend: Web/mobile UI with HTML/CSS/JS (or framework).
o Backend: RESTful APIs or server logic.
o Database: Schema design, SQL queries.
o Integration: API calls, authentication, error handling.

Prompt:-
Generate complete code for a full-stack application with frontend, backend, and database. Use simple technologies like HTML/CSS/JS and Node.js or similar. Include API integration, authentication, and error handling. Write clean, beginner-friendly and well-structured code.

Code:-

```
#  Task - 03
# Generate complete code for a full-stack application with frontend, backend, and database. Use simple technolo
"""# Due to the complexity of a full-stack application, I will provide a simplified version of an AI-assisted p
# Backend (Node.js with Express)
from numpy import require
const express = require('express');
const mongoose = require('mongoose');
const app = express();
app.use(express.json());
// Connect to MongoDB
mongoose.connect('mongodb://localhost:27017/fitnessApp', { useNewUrlParser: true, useUnifiedTopology: true });
// Define User Schema
const userSchema = new mongoose.Schema({
    name: String,
    age: Number,
    weight: Number,
    height: Number,
    fitnessGoals: [String],
    preferredWorkoutTypes: [String]
});
const User = mongoose.model('User', userSchema);
// API Endpoint to create a new user
app.post('/api/users', async (req, res) => {
    try {
        const user = new User(req.body);
        await user.save();
        res.status(201).send(user);
    } catch (error) {
        res.status(400).send({ error: 'Error creating user' });
    }
});
// API Endpoint to get workout plan (simplified)
app.get('/api/workout-plan/:userId', async (req, res) => {
    try {
        const user = await User.findById(req.params.userId);
        if (!user) {
            return res.status(404).send({ error: 'User not found' });
        }
        // Simplified workout plan generation based on fitness goals
        let workoutPlan = [];
        if (user.fitnessGoals.includes('weight loss')) {
            workoutPlan.push('30 minutes of cardio');
        }
        if (user.fitnessGoals.includes('muscle gain')) {
            workoutPlan.push('Strength training 3 times a week');
        }
        res.send({ workoutPlan });
    } catch (error) {
        res.status(400).send({ error: 'Error fetching workout plan' });
    }
});
// Start the server
app.listen(3000, () => {
    console.log('Server is running on port 3000');
});
# Frontend (HTML/CSS/JS)
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>AI-Assisted Personal Fitness Trainer</title>
    <style>
        body { font-family: Arial, sans-serif; }
        .container { max-width: 600px; margin: 0 auto; padding: 20px; }
        input, button { display: block; margin: 10px 0; padding: 10px; width: 100%; }
```

```html
    </style>
</head>
<body>
    <div class="container">
        <h1>AI-Assisted Personal Fitness Trainer</h1>
        <input type="text" id="name" placeholder="Name">
        <input type="number" id="age" placeholder="Age">
        <input type="number" id="weight" placeholder="Weight (kg)">
        <input type="number" id="height" placeholder="Height (cm)">
        <input type="text" id="fitnessGoals" placeholder="Fitness Goals (comma separated)">
        <input type="text" id="preferredWorkoutTypes" placeholder="Preferred Workout Types (comma separated)">
        <button onclick="createUser()">Create User</button>
        <button onclick="getWorkoutPlan()">Get Workout Plan</button>
        <div id="workoutPlan"></div>
    </div>
    <script>
        async function createUser() {
            const user = {
                name: document.getElementById('name').value,
                age: parseInt(document.getElementById('age').value),
                weight: parseInt(document.getElementById('weight').value),
                height: parseInt(document.getElementById('height').value),
                fitnessGoals: document.getElementById('fitnessGoals').value.split(','),
                preferredWorkoutTypes: document.getElementById('preferredWorkoutTypes').value.split(',')
            };
            const response = await fetch('/api/users', {
                method: 'POST',
                headers: { 'Content-Type': 'application/json' },
                body: JSON.stringify(user)
            });
            const data = await response.json();
            alert('User created with ID: ' + data._id);
        }
        async function getWorkoutPlan() {
            const userId = prompt('Enter User ID to get workout plan');
            const response = await fetch(`/api/workout-plan/${userId}`);
            const data = await response.json();
            document.getElementById('workoutPlan').innerText = 'Recommended Workout Plan: ' + data.workoutPlan.join(', ');
        }
    </script>
</body>
</html>
# This code provides a basic structure for a full-stack application with a Node.js backend and a simple HTML/CSS/JS frontend.
```

Output:-

A web page with a form to create a user profile and a button to fetch a workout plan based on the user's fitness goals. When the user creates a profile, they will receive an alert with their user ID. When they click the button to get a workout plan, they will be prompted to enter their user ID, and the recommended workout plan will be displayed on the page.

Justification:

This code is designed to be simple and beginner-friendly, making it suitable for a college capstone project. It demonstrates the core functionalities of a full-stack application, including API integration, database interaction, and basic error handling. The use of Node.js and MongoDB allows for easy setup and scalability, while the HTML/CSS/JS frontend provides a straightforward user interface for interacting with the application.

Task 4 – Documentation & Code Quality

- **Generate automatic documentation and inline comments using AI.**
- **Ensure clear README.md for project setup and usage.**
- **Conduct AI-assisted code review to refine readability and maintainability.**

Prompt:-

Create proper documentation for the project including README.md. Explain setup steps, features, and usage clearly. Add meaningful comments in the code for readability. Ensure the content looks natural and easy to understand.

Code:-

```
# README.md
# AI-Assisted Personal Fitness Trainer Application
## Overview
The AI-Assisted Personal Fitness Trainer is a full-stack application designed to provide personalized workout plans and real-time guidance
## Features
- Personalized Workout Plans: Generate workout plans based on user input such as age, weight, height, fitness goals, and preferred workout
- Real-Time Guidance: Provide feedback during workouts to ensure proper form and technique (future enhancement).
- Progress Tracking: Track workout history and improvements in strength and endurance (future enhancement).
- Nutrition Recommendations: Offer personalized nutrition advice to complement workout plans (future enhancement).
## Technologies Used
- Backend: Node.js with Express
- Database: MongoDB
- Frontend: HTML, CSS, JavaScript
## Setup Instructions
1. Clone the repository:
```bash
git clone https://github.com/yourusername/ai-fitness-trainer.git
```
2. Install dependencies:
```bash
cd ai-fitness-trainer
npm install
```
3. Start the development server:
```bash
npm run dev
```
4. Open the frontend in a web browser:
```bash
open index.html
```
Usage
1. Fill in the user information (name, age, weight, height, fitness goals, and preferred workout types) in the form on the homepage.
2. Click the "Create User" button to create a user profile. An alert will display the user ID.
3. Click the "Get Workout Plan" button and enter the user ID to retrieve the recommended workout plan based on the user's fitness goals.
Future Enhancements
- Implement real-time guidance using computer vision to provide feedback on exercise form and technique.
- Add progress tracking features to monitor workout history and improvements over time.
- Integrate nutrition recommendations to provide personalized dietary advice based on user goals.
Conclusion
This project serves as a foundational implementation of an AI-assisted personal fitness trainer application. It demonstrates the core func
Backend Code (Node.js with Express)
```javascript
const express = require('express');
const mongoose = require('mongoose');
const app = express();
app.use(express.json());
// Connect to MongoDB
mongoose.connect('mongodb://localhost:27017/fitnessApp', { useNewUrlParser: true, useUnifiedTopology: true });
// Define User Schema
const userSchema = new mongoose.Schema({
    name: String,
    age: Number,
    weight: Number,
    height: Number,
    fitnessGoals: [String],
    preferredWorkoutTypes: [String]
});
const User = mongoose.model('User', userSchema);
// API Endpoint to create a new user
app.post('/api/users', async (req, res) => {
    try {
        const user = new User(req.body);
        await user.save();
        res.status(201).send(user);
// API Endpoint to get workout plan (simplified)
app.get('/api/workout-plan/:userId', async (req, res) => {
    try {
        const user = await User.findById(req.params.userId);
        if (!user) {
            return res.status(404).send({ error: 'User not found' });
        }
        // Simplified workout plan generation based on fitness goals
        let workoutPlan = [];
        if (user.fitnessGoals.includes('weight loss')) {
            workoutPlan.push('30 minutes of cardio');
        }
        if (user.fitnessGoals.includes('muscle gain')) {
            workoutPlan.push('Strength training 3 times a week');
        }
        res.send({ workoutPlan });
    } catch (error) {
        res.status(400).send({ error: 'Error fetching workout plan' });
    }
});
// Start the server
app.listen(3000, () => {
    console.log('Server is running on port 3000');
});
```
```

```
# Frontend Code (HTML/CSS/JS)
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>AI Fitness Trainer</title>
</head>
<body>

</body>
</html>
```
```javascript
    <style>
        body { font-family: Arial, sans-serif; }
        .container { max-width: 600px; margin: 0 auto; padding: 20px; }
        input, button { display: block; margin: 10px 0; padding: 10px; width: 100%; }
    </style>
    <div class="container
">
        <h1>AI-Assisted Personal Fitness Trainer</h1>
        <input type="text" id="name" placeholder="Name">
        <input type="number" id="age" placeholder="Age">
        <input type="number" id="weight" placeholder="Weight (kg)">
        <input type="number" id="height" placeholder="Height (cm)">
        <input type="text" id="fitnessGoals" placeholder="Fitness Goals (comma separated)">
        <input type="text" id="preferredWorkoutTypes" placeholder="Preferred Workout Types (comma separated)">
        <button onclick="createUser()">Create User</button>
        <button onclick="getWorkoutPlan()">Get Workout Plan</button>
        <div id="workoutPlan"></div>
    </div>
     <script>
         async function createUser() {
             const user = {
                 name: document.getElementById('name').value,
                 age: parseInt(document.getElementById('age').value),
                 weight: parseInt(document.getElementById('weight').value),
                 height: parseInt(document.getElementById('height').value),
                 fitnessGoals: document.getElementById('fitnessGoals').value.split(','),
                 preferredWorkoutTypes: document.getElementById('preferredWorkoutTypes').value.split(',')
             };
             const response = await fetch('/api/users', {
                 method: 'POST',
                 headers: { 'Content-Type': 'application/json' },
                 body: JSON.stringify(user)
             });
             const data = await response.json();
             alert('User created with ID: ' + data._id);
         }
         async function getWorkoutPlan() {
             const userId = prompt('Enter User ID to get workout plan');
             const response = await fetch(`/api/workout-plan/${userId}`);
             const data = await response.json();
             document.getElementById('workoutPlan').innerText = 'Recommended Workout Plan: ' + data.workoutPlan.join(', ');
         }
     </script>
 ```
 # Conclusion
 This documentation provides a comprehensive overview of the AI-Assisted Personal Fitness Trainer application, including setup instructions,

 """
```

Output:-

A clear and concise README.md file that explains the project, its features, setup instructions, and usage in a way that is easy for students to understand and follow. The code includes meaningful comments to enhance readability and comprehension.

Justification:

This documentation is essential for guiding users through the setup and usage of the application. It ensures that students can easily understand the purpose of the project, how to get it running, and how to use its features effectively. The comments in the code further enhance understanding by explaining the functionality of different sections, making it more accessible for beginners.

Task 5 – Testing & Debugging
• Use AI to:
o Generate unit tests.
o Identify vulnerabilities/security flaws.
o Suggest optimizations for performance.

Prompt:-
Generate unit tests and basic test cases for the application. Identify possible bugs, security issues, and improvements. Suggest performance optimizations in simple terms. Keep explanations practical and easy for students.

Code:-

```
# Unit Tests for Backend API Endpoints
const request = require('supertest');
const app = require('./app'); // Assuming the Express app is exported from app.js
const mongoose = require('mongoose');
const User = require('./models/User'); // Assuming User model is defined in models/User.js
const { MongoMemoryServer } = require('mongodb-memory-server');
let mongoServer;
beforeAll(async () => {
    mongoServer = await MongoMemoryServer.create();
    const uri = mongoServer.getUri();
    await mongoose.connect(uri, { useNewUrlParser: true, useUnifiedTopology: true });
});
afterAll(async () => {
    await mongoose.disconnect();
    await mongoServer.stop();
});
describe('POST /api/users', () => {
    it('should create a new user', async () => {
        const res = await request(app)
            .post('/api/users')
            .send({
                name: 'John Doe',
                age: 30,
                weight: 80,
                height: 180,
                fitnessGoals: ['weight loss'],
                preferredWorkoutTypes: ['cardio']
            });
        expect(res.statusCode).toEqual(201);
        expect(res.body).toHaveProperty('_id');
    });
});
describe('GET /api/workout-plan/:userId', () => {
    it('should return a workout plan for a valid user', async () => {
        const user = new User({
            name: 'Jane Doe',
            age: 25,
            weight: 65,
            height: 165,
            fitnessGoals: ['muscle gain'],
            preferredWorkoutTypes: ['strength training']
        });
        await user.save();
        const res = await request(app).get(`/api/workout-plan/${user._id}`);
        expect(res.statusCode).toEqual(200);
        expect(res.body.workoutPlan).toContain('Strength training 3 times a week');
    });
    it('should return 404 for an invalid user', async () => {
        const res = await request(app).get('/api/workout-plan/invalidUserId');
        expect(res.statusCode).toEqual(404);
    });
});
# Possible Bugs and Security Issues
1. Input Validation: The backend does not currently validate user input, which could lead to issues such as missing fields or invalid data
2. Authentication: The application lacks authentication, which means anyone can access the API endpoints. Implementing user authentication
3. Error Handling: The error handling in the backend is basic. More detailed error messages and logging can help identify and resolve issue
# Performance Optimizations
1. Caching: Implement caching strategies to reduce database queries and improve response times.
2. Database Indexing: Ensure that frequently queried fields are indexed to speed up data retrieval.
3. Asynchronous Operations: Use asynchronous operations where possible to avoid blocking the main thread and improve overall performance.
```

Output:-
- A set of unit tests that validate the functionality of the backend API endpoints, ensuring that user creation and workout plan retrieval work as expected.
- Identification of potential bugs and security issues, along with practical suggestions for improvements.

Task 6 – Deployment & Presentation
• Deploy application on a cloud/free hosting platform(GitHub Pages, Firebase, etc.).
• Prepare a final presentation including:
o Problem statement
o Solution architecture
o Live demo
o Reflection on AI assistance and limitations

Prompt:-
Provide steps to deploy the project on a free platform like Vercel or Firebase. Create a simple presentation with problem, solution, architecture, & demo. Include key points for explaining the project confidently. Make the content clear, natural, and human-like.

Code:-

```
# Task - 06
# Provide steps to deploy the project on a free platform like Vercel or Firebase. Create a simple presentation with problem, solution, arch
"""
# Deployment Steps for Vercel
1. Create a Vercel Account: Sign up for a free account on Vercel
2. Install Vercel CLI: Install the Vercel CLI globally on your machine using npm:
```bash
npm install -g vercel
```
3. Initialize Vercel in Your Project: Navigate to your project directory and run:
```bash
vercel init
```
4. Configure Vercel: Follow the prompts to configure your project. Choose the appropriate settings for your backend and frontend.
5. Deploy Your Project: Run the following command to deploy your project:
```bash
vercel --prod
```
Presentation Outline
1. Introduction
   - Briefly introduce the problem of personalized fitness training and the need for an AI-assisted solution.
2. Problem Statement
   - Explain the challenges individuals face in creating effective workout plans and maintaining motivation.
3. Solution Overview
   - Introduce the AI-Assisted Personal Fitness Trainer application as a solution to these challenges.
4. System Architecture
   - Describe the architecture of the application, including the frontend, backend, and database components.
5. Key Features
   - Highlight the main features of the application, such as personalized workout plans, real-time guidance, and progress tracking.
6. Demo
   - Provide a live demonstration of the application, showcasing how to create a user profile and retrieve a workout plan.
7. Conclusion
    - Summarize the benefits of the application and its potential impact on users' fitness journeys.
Key Points for Explanation
- Emphasize the personalized nature of the workout plans and how they are tailored to individual goals and preferences.
- Highlight the use of AI and machine learning in generating workout plans and providing real-time feedback.
- Discuss the potential for future enhancements, such as integrating computer vision for exercise form correction and adding nutrition reco
- Address the importance of user data security and the steps taken to protect user information.
- Conclude by reinforcing the value of the application in helping users achieve their fitness goals and improve their overall health.
```

Output:-
- A successful deployment of the application on Vercel, making it accessible to users.
- A clear and engaging presentation that effectively communicates the problem, solution, architecture, and demo of the AI-Assisted Personal Fitness Trainer application, allowing the presenter to confidently explain the project to an audience.

Justification:
This presentation outline and deployment steps are designed to help students effectively communicate their project while also ensuring that their application is accessible to users. The key points for explanation are crafted to highlight the unique features and benefits of the application, making it easier for students to explain their work confidently and clearly. The deployment steps provide a practical guide for students to get their application online, allowing them to share their project with a wider audience.